

NaviCore

Intelligent Maritime Operations Platform

NaviCore is a comprehensive maritime operations and intelligence platform that combines fleet management, cargo optimization, risk analysis, and geopolitical intelligence monitoring.

Core Features

1. Fleet & Voyage Management

- Vessel tracking (AT_PORT, IN_TRANSIT, MAINTENANCE status)
- Voyage planning with origin/destination ports
- Real-time Cost Accrual Engine (tracks fuel/crew costs every 5s via WebSocket)

2. Cargo Management & Optimization

- QR code-based cargo tracking and Hazard classification
- AI Cargo Optimizer for weight balancing, deck slot assignments, and stability scoring

3. Weather & Marine Conditions

- Real-time weather data (Open-Meteo) and marine forecasts (waves, swell)
- Interactive weather layer visualization on the 3D dashboard

4. Risk Analysis & Navigation (MCP)

- Multi-factor Risk Scoring (Weather, Fuel, Distance, Sea State)
- Optimal Fuel Stop Recommendations based on prices, fees, and distance

5. News Aggregation & AI Insights

- 60+ global RSS feeds with automatic Groq AI Threat Classification
- Geo-tagging, news clustering, and geopolitical hotspot detection

6. Crew & Port Utilities

- Crew shift tracking, zone management, and idempotency key safety
- Directory of 100+ global ports and Haversine nautical mile calculations

Applications Ecosystem

- Dashboard: Main operations center (React 19, Maplibre GL, Deck.gl)
- Deckhand App: Mobile QR scanner for operations (React, html5-qrcode)
- Vessel Node: Offline-first PWA for vessel crews (PWA, localForage)

Technology Stack

- Frontend: React 19, TypeScript, Tailwind CSS, Vite
- Backend: Node.js/Express 5, Prisma ORM, Socket.io

- Database & Caching: PostgreSQL, Redis (Upstash)
- External APIs: Open-Meteo, Groq AI